

Pratham Desai

BTech CSE Graduate



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EDUCATION:

BACHELOR OF TECHNOLOGY IN COMPUTER SCIENCE ENGINEERING

2022 - 2026
GSFC University

CLASS 12TH
2021 - 2022
Kendriya Vidyalaya No.2 E.M.E Baroda

CLASS 10TH
2019 - 2020
Kendriya Vidyalaya No.2 E.M.E Baroda

ABOUT ME:

I enjoy exploring how technology works beneath the surface. I like understanding problems, experimenting with ideas, and turning them into practical solutions. My curiosity drives continuous learning, while sharing what I learn strengthens my communication and problem-solving skills.

Languages: Gujarati (Native), Hindi, English

PROFESSIONAL SUMMARY

B.Tech Computer Science graduate with experience in cybersecurity, security engineering, backend, full-stack, systems programming, and real-time development. Proficient in Python, FastAPI, Flask, React.js, TypeScript, Node.js, Express.js, Rust, REST APIs, WebSockets, Server-Sent Events (SSE), PostgreSQL, SQLite, Docker, Podman, and Kubernetes. Experienced in enterprise security solutions involving cryptographic auditing, CBOM generation, vulnerability assessment, RBAC, JWT authentication, cybersecurity compliance, Post-Quantum Cryptography readiness, token security, and automated security validation. Exposure to concurrent backend systems, distributed/local-network applications, AI/ML, computer vision, desktop development, blockchain/Web3, and cloud-independent systems, building production-oriented applications across cybersecurity, automation, and enterprise software.

TECHNICAL EXPOSURE

- Programming:** Python, JavaScript
- Backend & Full Stack:** FastAPI, Flask, Node.js, Express.js, React.js, Next.js, REST APIs, WebSockets, SSE, Tailwind CSS, Material UI
- AI Tools & Prompt Engineering:** ChatGPT, Claude, Gemini Pro; Code, Image & Video Generation Prompting; AI-Assisted Development & Productivity
- Cybersecurity:** Cryptographic Auditing, CBOM, CycloneDX, cdxgen, Security Compliance, RBAC, JWT, OpenLDAP, Vulnerability Assessment, Network Forensics, Phishing Detection, Steganography, Threat Intelligence, IoT Security
- AI / ML & Computer Vision:** PyTorch, CUDA, OpenCV, InsightFace, ArcFace, DeepFace, FAISS, scikit-learn, YOLOv8, MTCNN, NumPy, Pandas
- Databases:** PostgreSQL, MySQL, MariaDB, MongoDB, SQLite, Sequelize
- Cloud & Containers:** Docker, Podman, Kubernetes, kubectl, CI/CD
- Others:** PyQt6, PySide6, Android, Solana/Anchor, Postman, Selenium, Nmap, Shodan, Scapy

EXPERIENCE

ACADEMIC ASSOCIATE : PYTHON PROGRAMMING

GSFC University, Vadodara, Gujarat

July 2025 – October 2025

- Conducted lab lectures for B.Tech CSE 3rd semester students covering Python fundamentals, variables, data types, operators, conditional statements, loops, functions, and basic problem-solving.
- Provided supplementary learning materials and practical exercises to strengthen programming concepts.
- Prepared students for end-semester practical examinations, focusing on programming logic and problem-solving.
- Provided individualized support to students and helped bridge the gap between theoretical concepts and practical implementation.
- Maintained attendance and academic progress records for approximately 28 students.

ACADEMIC ASSOCIATE : PYTHON PROGRAMMING

GSFC University, Vadodara, Gujarat

January 2025 - April 2025

- Conducted lab lectures for B.Tech FEHS 2nd semester students covering Python fundamentals, variables, data types, operators, conditional statements, loops, functions, and basic problem-solving.
- Provided supplementary learning materials and practical exercises to strengthen programming concepts.
- Prepared students for end-semester practical examinations, focusing on programming logic and problem-solving.
- Provided individualized support to students and helped bridge the gap between theoretical concepts and practical implementation.
- Maintained attendance and academic progress records for approximately 35 students.

ACADEMIC ASSOCIATE : C PROGRAMMING

GSFC University, Vadodara, Gujarat

August 2024 - November 2024

- Conducted C programming labs for B.Sc Data Science 1st-semester students, covering fundamentals, variables, data types, operators, conditional statements, loops, functions, arrays, and basic problem-solving.
- Developed supplementary learning materials and practical exercises to strengthen programming concepts.
- Prepared students for end-semester practical examinations, focusing on programming logic and problem-solving.
- Provided individualized academic support and helped students connect theoretical concepts with practical implementation.
- Maintained attendance and academic progress records for approximately 45 students.

INTERSHIPS

CYBERSECURITY INTERN

(8TH SEM)

Tata Consultancy Services (TCS) – Garima Park, Gandhinagar, Gujarat

05 February 2026 – 05 June 2026

- Developed CipherLens, a security platform for generating and analyzing Cryptographic Bills of Materials (CBOM) across source code, Git repositories, container images, and Kubernetes environments.
- Built high-performance security scanning and validation workflows, combining parallel processing, cryptographic analysis, compliance assessment, and automated risk evaluation.
- Contributed to enterprise-grade backend, database, authentication, and dashboard components, supporting real-time scan monitoring and security reporting.
- Worked within enterprise development processes including access provisioning, RFC/ticket management, environment configuration, software approvals, and corporate governance.

Project: CipherLens – Cryptographic Bill of Materials (CBOM) Analyser

Technologies: Python, FastAPI, React.js, PostgreSQL, SQLite, Podman, Kubernetes, CycloneDX, cdxgen, WebSockets, SSE, OpenLDAP, JWT

- Developed and evolved CipherLens, a cryptographic auditing platform for generating and analyzing CBOMs across source code, Git repositories, Docker/OCI container images, and Kubernetes workloads.
- Engineered high-performance scanning pipelines using parallel processing, ProcessPoolExecutor, IPC, CycloneDX/cdxgen, and custom regex analysis for efficient cryptographic asset discovery.
- Implemented security validation against NIST SP 800-131A, CERT-In, and ETSI, including risk scoring and Post-Quantum Cryptography (PQC) readiness assessment.
- Built secure enterprise architecture with FastAPI, PostgreSQL, OpenLDAP, JWT/RBAC, alongside real-time scan monitoring using WebSockets and SSE.
- Extended the platform with an offline CLI for CI/CD and air-gapped environments and a rootless Podman/Kubernetes scanning pipeline for containerized workloads.
- Developed automated reporting and persistence workflows supporting CycloneDX JSON, HTML, PDF, and CSV outputs.

PYTHON INTERN

(7TH SEM)

PayGURU Technosoft Pvt. Ltd., Vadodara, Gujarat

01 December 2025 – 10 January 2026

- Developed and contributed to four production-oriented systems spanning visitor management, WhatsApp automation, computer vision, and IoT security.
- Built scalable backend and automation workflows using Python, Node.js, Flask, REST APIs, multi-threading, and database systems.
- Developed real-time solutions integrating CCTV/RTSP streams, WhatsApp webhooks, email notifications, and IoT device APIs.
- Applied performance optimization, secure credential management, automation, and database persistence to address real-world business and operational requirements.

Reference:

Mr. Ryan Mathew - Project Lead at PayGURU - +91 98795 39606

INTERSHIPS

Projects:

PookiePass : Visitor Management System

Technologies: React.js, Node.js, Express.js, MariaDB, Sequelize, REST APIs, Nodemailer, Multi-Tenancy

- Developed a SaaS-ready multi-tenant Visitor Management System with centralized deployment and strict tenant-level data isolation.
- Implemented real-time visitor approval workflows, automated email/WhatsApp notifications, and a polling-based waiting room.
- Built dynamic white-labeling that adapts branding and themes based on the tenant's company URL.

WhatsApp Automation Bot : PayGURU Marketing

Technologies: Python, Flask, Meta Graph API, SQLite, Multi-threading, PyQt, JSON

- Developed a WhatsApp Business CRM and marketing automation system for automated product showcases, customer engagement, and support workflows.
- Built configuration-driven messaging and bulk contact processing to support large-scale campaigns.
- Implemented multi-threaded webhooks, media caching, and a PyQt event bridge for real-time lead synchronization, reducing API latency by approximately 40%.

CCTV-Based Real-Time Face Recognition & Attendance System

Technologies: Python, OpenCV, PyTorch (CUDA), InsightFace, ArcFace, MTCNN, FAISS, RTSP/ISAPI, NumPy, Pandas, Albumentations

- Developed a GPU-accelerated face recognition and attendance system using live CCTV feeds and Hikvision IP cameras.
- Integrated FAISS vector search for scalable face-embedding matching and engineered low-latency RTSP processing using frame skipping, buffering, and multi-threading.
- Implemented enrollment validation, image augmentation, identity stabilization, and automated attendance logging to reduce duplicate entries and false positives.

Session-Based Access Control – Python Security Utility

Technologies: Python, Multi-threading, REST APIs (ISAPI), MySQL/MariaDB, Selenium, AES Encryption

- Developed a security automation utility for Hikvision IoT device discovery, credential management, and automated password rotation.
- Engineered a 50+ thread network scanner using MAC/Serial fingerprinting to continuously track devices despite IP changes.
- Implemented encrypted credential storage, automated ISAPI-based password rotation, and Selenium fallback automation for legacy firmware compatibility.

INTERSHIPS

PYTHON INTERN

(6TH SEM)

PayGURU Technosoft Pvt. Ltd., Vadodara, Gujarat

09 June 2025 – 04 July 2025

- Collaborated in a two-member team to develop and deploy PookieTrak, a cross-platform desktop solution for real-time software and network-device monitoring.
- Developed PyQt6-based interfaces, multithreaded monitoring workflows, automated reporting, and email notification systems.
- Performed testing, debugging, performance optimization, and documentation across Windows and Linux, gaining hands-on exposure to the software development lifecycle.

Project: PookieTrak – Tracking & Alert System

Technologies: Python, Multi-threading, REST APIs (ISAPI), MySQL/MariaDB, Selenium, AES Encryption

- Developed a real-time desktop monitoring solution for software usage and network-device availability, with automated logging and reporting.
- Engineered a 50+ thread network scanner using MAC/Serial fingerprinting and implemented automated device tracking despite changing IP addresses.
- Built secure credential management and password rotation using ISAPI and encrypted storage, with database synchronization.
- Implemented reliable infrastructure alerting with SMTP-based notifications, HTML reporting, and auto-recovery, alongside Selenium fallback automation for legacy devices.

MERN STACK INTERN

(5TH SEM)

Sannibh Technologies, Vadodara, Gujarat

02 December 2024 – 03 January 2025

- Collaborated in a two-person team to develop Brass ERP, a full-stack ERP solution for brass-part manufacturing.
- Contributed to backend APIs, authentication, database integration, and responsive frontend modules across business workflows.
- Worked in a hybrid development environment, independently delivering features while coordinating with teammates and stakeholders.

Project: Brass ERP – Manufacturing ERP System

Technologies: MySQL, Express.js, React.js, Node.js, Sequelize, REST APIs, JWT, Zustand, Tailwind CSS, Vite

- Developed a modular ERP application for brass-part manufacturing covering client management, quotations, production, sales, and raw materials.
- Implemented JWT-based authentication, protected API routes, RESTful services, input validation, and cookie-based session management.
- Built responsive React interfaces with Tailwind CSS and Zustand for efficient state management across modules.
- Structured the backend using MVC architecture and Sequelize ORM, with reusable components, error handling, and maintainable code.

PERSONAL PROJECTS

CINEMATIC PARALLAX SINGLE-PAGE WEB APPLICATION

26 June 2026

Technologies: React.js, JavaScript (ES6+), Vite, Sass (SCSS), HTML5, CSS3, Web3Forms API, HTML5 Audio API

Libraries Used: Framer Motion, React Icons, EmailJS

- Developed an interactive, high-performance portfolio website featuring rich physics-based animations and responsive design.
- Engineered scroll-linked Parallax effects, multi-layer depth transitions, and entrance animations utilizing Framer Motion.
- Built a modular frontend architecture with React 18 and Sass (SCSS), incorporating a custom-tracked animated cursor and full-screen CSS Scroll-Snap navigation.
- Implemented an asynchronous client-side contact submission flow with REST API integration, optimized with Vite for fast bundling and minimal load latency.

HOMEVAULT : CROSS-DEVICE WIRELESS FILE SHARING & LOCAL LINK APP

30 June 2026

Technologies: Rust, TypeScript, React.js, Tauri v2, SQLite, Node.js, Vite, HTML5, CSS3, TailwindCSS, mDNS (Zeroconf), UDP Sockets, HTTP/REST, Server-Sent Events (SSE), Wake-on-LAN (WoL)

Libraries used: Axum, Tokio, Rusqlite, Tower-HTTP, Notify, mDNS-SD, Wakey, Image, Zip, Sysinfo, Request, Serde, Infer, Walkdir, Tracing, Directories, UUID, Zustand, TanStack Query, Material UI (MUI), React Virtuoso, Video.js, React Photo View, React Doc Viewer, React Selecto, Axios, QRCode.react, Emotion, PostCSS

- Built a cross-platform, zero-cloud wireless file sharing and device-linking desktop application engineered with Tauri v2, Rust (Axum), and React + TypeScript, enabling seamless laptop-to-laptop and phone-to-PC file transfer and browsing over local Wi-Fi.
- Architected a zero-configuration peer discovery protocol using mDNS-SD (Zeroconf) and fallback UDP broadcast sockets (port 8001) for instant device pairing, device handshake negotiation, and automated Wake-on-LAN (WoL via Wakey) magic packet signaling.
- Engineered secure, PIN-authenticated peer sessions and high-throughput multipart file transfer pipelines supporting recursive folder preservation, chunked stream uploads, and atomic rename file integrity.
- Implemented low-latency HTTP 206 Partial Content (byte-range) streaming in Rust for immediate random-access media seeking, combined with an event-driven background filesystem indexer via Notify OS hooks, Server-Sent Events (SSE), and SQLite WAL-mode (Rusqlite).
- Designed a responsive client dashboard with React Virtuoso, Zustand, and Video.js, delivering 60 FPS browsing across 10,000+ files, drag-and-drop transfers, in-browser document previews, and QR code mobile pairing.

PERSONAL PROJECTS

JUDGE ME IF YOU CAN : REAL-TIME MULTIPLAYER WEB GAME

27 July 2026

Technologies: TypeScript, Node.js, Express.js, WebSockets, React.js, Vite, TailwindCSS, HTML5, CSS3, NPM Workspaces

Libraries used: Zustand, Framer Motion, Phaser 3, Howler.js, Canvas-Confetti, Lucide React, React Router DOM, ws, NanoID, CORS

- Developed a full-stack, real-time interactive multiplayer party web application engineered as a high-performance modular TypeScript monorepo with an event-driven WebSocket communication layer for synchronized room lobbies, PIN authentication, and zero-latency state transitions.
- Engineered multi-phase game engines with complex rulesets, featuring live vote broadcasting, dynamic score-stealing, player elimination/ghosting mechanics, team revival gambles, and synchronized audio-visual feedback using Framer Motion and Howler.js.
- Built a real-time host administration and live telemetry dashboard with Zustand to monitor connected client payloads, trigger game flow overrides, and generate automated post-game analytics reports.

SHAWWTIE : FULL-STACK URL SHORTENER & ANALYTICS

05 August 2026

Technologies: JavaScript (ES6+), React 19, Node.js, Express.js, PostgreSQL, SQL, Neon DB, Vite, CSS3, REST APIs

Libraries used: Nano ID (nanoid), pg (node-postgres), CORS, Dotenv, Oxlint

- Engineered a full-stack URL shortening and click telemetry platform with a decoupled React SPA and modular Express REST API, utilizing Nano ID for collision-resistant 7-character base64 keys.
- Architected sub-millisecond redirection via HTTP 302 status codes to avoid browser cache poisoning, coupled with an asynchronous, non-blocking click-logging pipeline capturing native INET IPs and referrers into indexed PostgreSQL schemas.
- Built an analytics engine executing parallel Promise.all queries for 24h trends, 30-day click histories, and top traffic sources, integrated with a custom React 19 dashboard featuring local storage history synchronization.

GATEKEEP : AUTHENTICATION & SECURITY SERVICE

10 August 2026

Technologies: JavaScript (ES6+), React 19, Node.js, Express.js, PostgreSQL, Neon DB, Vite, CSS3, REST APIs, Oxlint

Libraries Used: jsonwebtoken (JWT), bcrypt, express-rate-limit, pg (node-postgres), cookie-parser, dotenv, zxcvbn

- Architected a production-ready authentication microservice pairing a decoupled React 19 SPA with a hardened Express REST API, enforcing 15-minute access tokens and 7-day single-use refresh tokens with zero raw tokens stored in persistent memory.
- Implemented automatic token rotation with replay attack detection and cascading revocations, hardening endpoints via SHA-256 database token hashing, constant-time bcrypt verification to prevent user enumeration, IP rate-limiting, and strict HttpOnly/SameSite cookies.
- Integrated client-side zero-knowledge security using the Have I Been Pwned API with Web Crypto SHA-1 \$k\$-Anonymity range queries and zxcvbn entropy evaluation, proactively blocking leaked or weak credentials before submission over an indexed PostgreSQL backend.

COLLEGE PROJECTS

NETRA-AI : AI-POWERED RETINAL CARDIOVASCULAR RISK PREDICTION SYSTEM (8TH SEM)

30 April 2026

Technologies: Python, PyTorch, FastAPI, Uvicorn, React.js, TypeScript, TailwindCSS, HTML5, CSS3, SQLite, PostgreSQL, CUDA, Git, EfficientNet (B0 / V2-S), ConvNeXt-Small, TorchVision, OpenCV, Pillow, Albumentations, NumPy, Pandas, SciPy, Scikit-learn, Matplotlib, Seaborn, SHAP, ReportLab, SQLAlchemy, Alembic, Pydantic v2, pydantic-settings, python-jose, passlib/bcrypt, slowapi, aiofiles, python-multipart, python-dotenv, google-genai (Gemini 2.5 Flash), React Router DOM

- Designed and trained a 3-class deep-learning pipeline for cardiovascular risk prediction (Low / Medium / High) from retinal fundus images using EfficientNetV2-S, 86.4% validation accuracy at epoch 13/40.
- Trained on a merged 19,466-image dataset (RFMiD + BRSET) with circular FOV cropping, CLAHE in LAB colour space, and Albumentations augmentation.
- Implemented AdamW, Label Smoothing, Focal Loss, cosine annealing with warm-up, gradient clipping, and WeightedRandomSampler to improve training and address class imbalance.
- Built a production-grade FastAPI asynchronous REST backend with a background inference job queue, three-tier RBAC (Admin / Technician / Receptionist), OAuth2 Bearer JWT authentication, rate-limited endpoints, magic-byte file validation, and HTTP audit logging.
- Developed a 7-section clinical intake workflow and integrated an inference pipeline using 30-sample Monte Carlo Dropout, uncertainty quantification, temperature-scaled softmax, and Test-Time Augmentation.
- Implemented model explainability using Grad-CAM++, SmoothGrad Integrated Gradients, and SHAP to generate per-scan visual heatmaps.
- Developed a React 19 + TypeScript SPA with role-based protected routes, a multi-step clinical intake wizard, and a live appointment queue system.
- Automated 8-section clinical PDF reports using ReportLab, including Grad-CAM overlays and probability charts, with on-demand SMTP/TLS email delivery.
- Integrated Gemini 2.5 Flash to synthesize de-identified patient vitals, blood pressure, lifestyle data, and dual-eye model probabilities into structured clinical risk narratives with contributing factors and actionable recommendations.

MAYABHEDI: THE ONE WHO SEES THROUGH DECEPTIONS (7TH SEM)

07 November 2025

Technologies: Python, PySide6, AES-256-GCM, Argon2id, scikit-learn, Nmap, Shodan API, Scapy, YOLOv8, OpenCV

- Developed a multi-domain cybersecurity desktop application integrating password security, cryptographic vaults, steganography, phishing detection, vulnerability assessment, network forensics, and physical-security monitoring.
- Implemented secure credential and file protection using AES-256-GCM, Argon2id, password-strength analysis, breach monitoring, authenticated encryption, and steganography.
- Built security analysis modules using scikit-learn, Nmap, Shodan API, and Scapy for phishing detection, vulnerability assessment, network monitoring, and threat intelligence.
- Trained and deployed a YOLOv8 computer-vision model for hidden-camera detection and developed a modular, multi-threaded PySide6 desktop interface.

COLLEGE PROJECTS

SCANSAGE: ALL-IN-ONE DOCUMENT UTILITY

(7TH SEM)

07 October 2025

Technologies: Java, Android, OpenCV, CameraX, ML Kit, iText7, Apache POI, Material UI

- Developed a Java-based Android document utility supporting PDF/image merging, splitting, conversion, compression, scanning, and format transformation.
- Implemented document processing using iText7, Apache POI, and PdfRenderer, with asynchronous processing to maintain responsive UI performance.
- Built a camera-based document scanner using OpenCV and CameraX, implementing edge detection, contour analysis, perspective correction, and image enhancement.
- Integrated ML Kit Selfie Segmentation for background removal and developed a customizable text-to-handwriting generator with high-resolution A4 output.

BLOCKBEATS : BLOCKCHAIN MUSIC PLAYER

(6TH SEM)

03 May 2025

Technologies: Rust, Anchor, Solana, Next.js, React.js, Tailwind CSS, Phantom Wallet, Arweave, Cloudinary, QuickNode, Axios

- Developed a Solana-based decentralized music platform focused on low-cost music publishing and distribution for emerging artists.
- Engineered Rust/Anchor smart contracts on Solana and integrated Phantom Wallet for Web3 authentication and blockchain interactions.
- Built a responsive Next.js/React.js frontend with Tailwind CSS and integrated Arweave/Cloudinary for decentralized and direct media storage.
- Implemented blockchain communication through QuickNode RPC and Axios, with Postman-based API testing and debugging.